

Transient Dynamics and Frequency-Dependent Impedance in RC, RL, LC, and LCR Circuits

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Abstract

In this lab I tested whether simple RC, RL, and LC circuits built from measured components behave as predicted by their textbook transient and impedance models. Using a digital oscilloscope, I recorded time-domain responses to step-like inputs (slow battery discharge and fast square-wave edges) and frequency-domain responses to sinusoidal driving, then fitted exponential and damped-sinusoid models in time and standard impedance/phase formulas in frequency using least-squares and χ^2 statistics. For the slow and fast RC transients, the fitted time constants ($\tau_C \approx 0.34$ s in Exercise 1 and $\tau_{RC} \approx 0.589$ ms in Exercise 2) agreed well with the values inferred from the measured R and C once the oscilloscope's 1 M Ω input was included as a parallel load. In contrast, the RL and LC transients yielded effective time constants that implied larger total series resistances and different damping for V_C and V_L , indicating that inductor losses and parasitic elements are important. In the frequency-domain measurements of Exercise 3, fitting the RC data gave $C_\phi \approx 21.4$ nF and $R_Z \approx 513$ Ω in good agreement with the bench values, while the RL impedance fit gave $L_Z \approx 43$ mH and $R_Z \approx 533$ Ω but the RL phase fit returned a smaller inductance ($L_\phi \approx 22$ mH), showing that amplitude data followed the simple model more closely than the phase data. Overall, the analysis demonstrates that RC behaviour is captured very accurately by the simple models when loading is accounted for, whereas RL and LC circuits show noticeable systematic deviations linked to additional resistance and non-ideal inductor properties.

1 Introduction

In this experiment, we study circuits comprising resistors, capacitors, and inductors. We observe the voltage response in each of these combinations to derive time constants and impedance values. The first two experiments use DC current to quantify time constants, with a notable difference in the experimental setup: where the first experiment uses a manual switch, while the second uses an automated switch. Moreover, the third experiment employs an AC current to measure impedance.

The equations for the voltage in the three different components is:

$$V_R(t) = RI(t) \quad (1)$$

for the resistor, with R being the resistance and I(t) the current in the resistor,

$$V_L(t) = L \frac{dI(t)}{dt} \quad (2)$$

for the Inductor, with L being the inductance and I(t) the current in the inductor

$$V_C(t) = \frac{1}{C} \int_0^t I(t') dt' \quad (3)$$

for the Capacitor, with C being the capacitance and I(t) being the current in the Capacitor.

1.1 DC circuits

Using these three equations we can find ODEs for the 3 different circuits that will be built, the LR circuit, the LC circuit, and the RC circuit. The ODEs are different when the source of voltage is on vs off, with a "driving term" on the right when the source is on, and a zero term when the source is off.

The equations of the charge voltage and current in each circuit when the voltage is switched on are:

with V_0 being the applied voltage. and the equations that are found when the voltage source is switched off are:

1.2 AC circuits and Impedance

While in DC circuits the voltage (and current) supplied are constant, in an AC circuit they are varying in time (oscillatory). This means we need to consider the impedance of a circuit instead of the resistance, which includes both a real and imaginary part. This is because in AC circuits voltage

Table 1: Circuit Behavior During Switch-On and Long-Time Steady State

Circuit	Charging (after switch-on)	Long Time
RC	$v_C(t) = V_0 (1 - e^{-t/(RC)})$ $i(t) = \frac{V_0}{R} e^{-t/(RC)}$	$v_C(\infty) = V_0$ $i(\infty) = 0$
RL	$i(t) = \frac{V_0}{R} (1 - e^{-Rt/L})$ $v_L(t) = V_0 e^{-Rt/L}$	$i(\infty) = \frac{V_0}{R}$ $v_L(\infty) = 0$
LC	$v_C(t) = V_0 (1 - \cos(\omega_0 t))$ $i(t) = CV_0 \omega_0 \sin(\omega_0 t)$ $\omega_0 = \frac{1}{\sqrt{LC}}$	No DC steady state (oscillatory)

Table 2: Circuit Behavior After Switch-Off (Discharging)

Circuit	Discharging (switch-off)
RC	$v_C(t) = V_0 e^{-t/(RC)}$ $i(t) = -\frac{V_0}{R} e^{-t/(RC)}$
RL	$i(t) = I_0 e^{-Rt/L}$ $v_L(t) = RI_0 e^{-Rt/L}$
LC	$q(t) = Q_0 \cos(\omega_0 t) + \frac{I_0}{\omega_0} \sin(\omega_0 t)$ $i(t) = -Q_0 \omega_0 \sin(\omega_0 t) + I_0 \cos(\omega_0 t)$ $\omega_0 = \frac{1}{\sqrt{LC}}$

and current oscillate at an angular frequency $\omega = 2\pi f$ with a phase difference ϕ with equations:

$$I(t) = I_0 \sin(\omega t) \quad (4)$$

$$V(t) = V_0 \sin(\omega t + \phi) \quad (5)$$

If we instead write each as complex numbers using phasors, with equations:

$$I(t) = I_0 e^{i(\omega t)} \quad (6)$$

$$V(t) = V_0 e^{i(\omega t + \phi)} \quad (7)$$

then we can now define impedance as:

$$Z = V/I = Re^{i\phi} \quad (8)$$

and it carries the same units as resistance (ohms).

You can find the impedance of any circuit component by replacing I in the equations with equation (6) and doing any operations needed. For a resistor the equation is already $V = IR$ and so the impedance of a resistor is:

$$Z_R = V/I = R \quad (9)$$

For an Inductor, we have the equations:

$$V = L \frac{dI}{dt} = L \frac{d}{dt} (I_0 e^{i(\omega t)}) = i\omega LI \quad (10)$$

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rearranging, we get:

$$Z = i\omega L \quad (11)$$

and for a capacitor, we have the equation:

$$V = \frac{1}{C} \int I(t) dt = \frac{1}{C} \int I_0 e^{i(\omega t)} dt = \frac{1}{i\omega C} I \quad (12)$$

rearranging, we get:

$$Z_C = \frac{1}{i\omega C} \quad (13)$$

Since the impedance of a circuit adds together in the same way as resistance does, the impedance of an LCR circuit is:

$$Z_{LCR} = R + i(\omega L - \frac{1}{\omega C}) \quad (14)$$

You can also find the phase difference by using both parts of the complex number, with equation:

$$\phi = \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right) \quad (15)$$

2 Methods and Procedures

2.1 Exercise 1

In this experiment, using an Oscilloscope(Keysight DSOX-1202G), 1 μF Capacitor, 470 k Ω resistor, a 1.5V Voltage battery and a switch, we built the RC circuit shown in Fig.11 in the appendix. We also used the oscilloscope channel 1 and 2 to measure V and V_C . Then we measured the capacitance, and resistance of the components, as seen in the appendix3:

The oscilloscope time range should then be increased to 2 seconds per division and both channel measurements started, and then we pressed the switch down, held it down for 5 seconds, and then let it go. The stop button on the oscilloscope was then used to freeze the data on the screen and check if the time constant was close to the theoretical value, and then the data was saved to a usb.

2.2 Exercise 2

To study a similar phenomenon in faster circuits, we instead built the circuit as seen in Fig.12 in the appendix, and use a square wave on the signal generator (on the oscilloscope) instead of the battery.

For the RC circuit, we used a 22nF capacitor as the "Z" device, 0.950kHz, 6.7V and 6.0V offset, and 28 k Ω resistor; and then measured the applied voltage V and the Voltage across the resistor V_R and saved the data to the usb. For the RL circuit, we used 505 Ω resistor and used the inductor as the "Z" device. We then measured the Applied Voltage V and the Voltage across the resistor V_R . For the LC circuit, we changed the frequency to 310 Hz, and the amplitude to 17pp, then measured V_C and V_L using CH1 and CH2 of the oscilloscope, and saved the data of each device to 2 separate files on the USB.

2.3 Exercise 3

To study what happens with AC currents, we instead built the circuit as seen in Fig. 13 in the appendix, and use a sin wave on the signal generator (on the oscilloscope) instead of the square wave.

For the RC circuit, we used the same resistor and capacitor (as the "Z" device in the circuit) as in Exercise 2, but then used 8.7Vpp, and no offset on the signal generator, and measured both the applied voltage V, the resistors voltage V_R , and the phase between both voltages ϕ for a range of different frequencies, from 1kHz to 30kHz going up in 0.5kHz increments.

For the RL circuit, we used a 505 Ω resistor and a 0.041 H inductor (as the "Z" device in the circuit). We used a 6Vpp

sin wave and no offset on the signal generator. We measured both the applied voltage V, the Inductor's voltage V_L , and the phase between both voltages ϕ for a range of different frequencies, from 25Hz to 3.2kHz going up in varying increments. We used 25Hz, then from 50Hz to 1kHz we went up in 50 Hz increments, then from 1kHz to 2kHz we went up in 200 Hz increments, then from 2kHz to 3.2kHz we went up in 400 Hz increments.

3 Analysis and Discussion

For this experiment we investigated the transient response of a RC circuit when a charged capacitor was allowed to discharge through a resistor after a step change in applied voltage. Representative traces and their fits are shown in Figure 1 (V_C) and Figure 2 (V_R). For each point we assigned a vertical uncertainty by combining (i) half of the least significant vertical division (LSB) and (ii) the vertical gain uncertainties quoted in the oscilloscope manual. These uncertainties appear as the error bars in both the voltage and residual plots. Horizontal (time-base) uncertainties are much smaller than the decay time and were neglected in the fit.

First, we Fit both $V_C(t)$ and $V_R(t)$ to the model 19 using the voltage uncertainties from the oscilloscope to obtain best-fit values of τ and their statistical uncertainties. Then compare the fitted time constants with the theoretical value $\tau_{th} = RC$ obtained from the measured component values, and interpret any discrepancies in terms of the resistance and capacitance in the circuit. Finally, using the chi-squared statistic and residual plots to test how well the exponential model describes the data and to reveal deviations from the ideal RC model.

3.1 Capacitor Voltage: $V_C(t)$

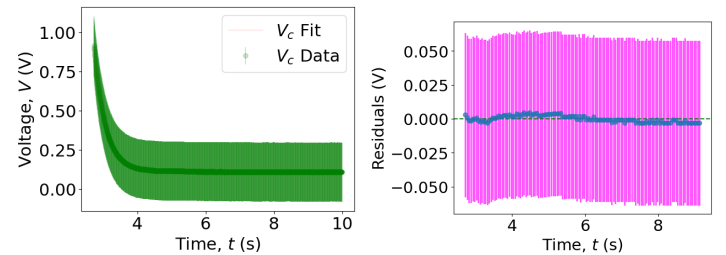


Figure 1: Left) Measured voltage across the capacitor V_C in the RC circuit during discharge following a step change in applied voltage, fitted with the exponential decay model. The best-fit time constant is $\tau = (0.34 \pm 0.03)$ s. The fit yields a chi-squared statistic of $\chi^2 = 28.5$, corresponding to a reduced chi-squared of $\chi^2_v = 0.837$ and a chi-squared probability $p = 0.735$. **Right)** Residuals of the fit uncertainty bars

Fitting the capacitor discharge data in Figure 1 to 19, yields

$$\tau_C = (0.342 \pm 0.003) \text{ s},$$

with chi-squared statistic $\chi^2 = 28.5$, reduced chi-squared $\chi^2_v = 0.837$, and chi-squared probability $p = 0.735$.

A reduced χ^2 slightly below one and a large p -value shows that the exponential model is consistent with the data and that the assigned uncertainties are slightly over-estimated. The residuals (lower panel of Figure 1) show no strong remaining time dependence: over the plotted range ($t \geq 3$ s) they are small compared to the uncertainty bars and are roughly symmetrically distributed about a nearly constant offset slightly below zero. This pattern indicates that a single-exponential decay provides a good description of the shape of $V_C(t)$ at long times, while the small offset likely reflects an unmodelled systematic effect such as a constant DC offset in the oscilloscope or a slight error in the zero-voltage baseline.

Thus the measured τ_C is about 30% smaller than RC , a discrepancy larger than the quoted uncertainties. This shows that the RC model is incomplete.

However, according to 2, the oscilloscope's vertical input has an input impedance of $\sim 1.0\text{ M}\Omega$ which is ran in parallel with the circuit resistor. So the effective resistance becomes

$$R_{\text{parallel}} = \left(\frac{1}{R} + \frac{1}{R_{\text{scope}}} \right)^{-1} \approx 3.18 \times 10^5 \Omega = 318\text{ k}\Omega.$$

This predicts a modified time constant

$$\tau_{\text{pred,loaded}} = R_{\text{parallel}}C \approx 0.333\text{ s},$$

about 2.8% smaller than the measured τ_C . Inverting this relationship to solve for capacitance gives

$$C_{\text{eff,C}} = \frac{\tau_C}{R_{\text{parallel}}} \approx 1.08 \times 10^{-6}\text{ F} = 1.08\text{ }\mu\text{F},$$

which differs slightly from the measured $C = 1.05\text{ }\mu\text{F}$ by only $\sim 2.8\%$. Given the approximations made for the additional input resistance this is a reasonable level of agreement.

Overall, the capacitor data show that the circuit behaves like an RC transient once the loading of the oscilloscope is included. Without that effect the experimental time constant would disagree strongly with typical RC behaviour.

Resistor Voltage $V_R(t)$

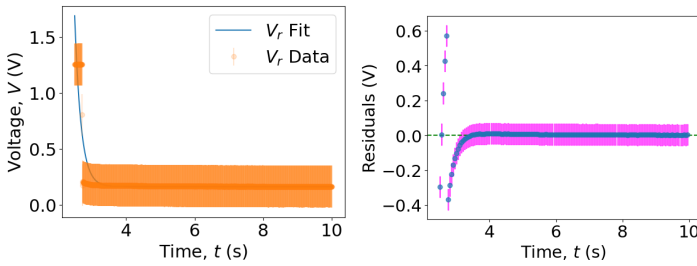


Figure 2: Left) Measured voltage across the resistor V_R in the RC circuit during discharge following a step change in applied voltage, fitted with the exponential decay model. The best-fit time constant is $\tau = (0.18 \pm 0.03)\text{ s}$. The fit yields a chi-squared statistic of $\chi^2 = 67.9$, corresponding to a reduced chi-squared of $\chi^2_{\nu} = 1.21$ and a chi-squared probability $p = 0.132$. **Right)** Residuals of the fit with uncertainty bars.

Applying the same model 19 to the resistor data in Figure 2, gives

$$\tau_R = (0.18 \pm 0.03)\text{ s},$$

with $\chi^2 = 67.9$, $\chi^2_{\nu} = 1.21$, and $p = 0.132$. Here the reduced χ^2 is slightly above one and the p -value is smaller (~ 0.13), which indicates the exponential model is still statistically acceptable, but the agreement is clearly poorer than $V_C(t)$.

The residuals in the lower panel of Figure 2 show that the earliest points deviate systematically from zero, but for $t \geq 3.5\text{ s}$ the residuals are small and roughly symmetrically scattered about the zero line. This pattern suggests that the single-exponential model describes the long-time behaviour of $V_R(t)$ well, while the first few points are affected by small unmodelled systematics such as an imperfect initial discharge or a slight DC offset in the oscilloscope.

Interpreting τ_R in the same way as before gives

$$R_{\text{eff,R}} = \frac{\tau_R}{C} \approx \frac{0.1844\text{ s}}{1.047 \times 10^{-6}\text{ F}} = (180 \pm 30)\text{ k}\Omega,$$

which is only about 38% of the nominal resistor value and roughly half of the effective resistance inferred from the capacitor data. Even if we include the scope loading and interpret the result as an effective capacitance using $R_{\text{parallel}} = 318\text{ k}\Omega$,

$$C_{\text{eff,R}} \approx \frac{\tau_R}{R_{\text{parallel}}} \approx 0.58\text{ }\mu\text{F},$$

which is about 45% smaller than the independent capacitance measurement.

Thus, while $V_R(t)$ clearly decays in time and is roughly compatible with an exponential, the extracted time constant is significantly shorter than both the RC value and the more realistic loaded value inferred from $V_C(t)$. The larger reduced χ^2 and residual structure mean that additional non-idealities (switching behaviour, offsets, extra leakage paths) are possibly affecting the resistor measurement more strongly. Because the fitting model includes a free offset V_{∞} , any small drift or offset can bias the value of τ_R downward, so we would say the time constant from V_C as more reliable.

Using only the measured component values,

$$\tau_{\text{th}} = (0.487 \pm 0.001)\text{ s},$$

so the deviations are $\frac{\tau_C - \tau_{\text{th}}}{\tau_{\text{th}}} \approx -30\%$, $\frac{\tau_R - \tau_{\text{th}}}{\tau_{\text{th}}} \approx -62\%$. Including the oscilloscope loading changes the prediction to $\tau_{\text{pred,loaded}} \approx 0.333\text{ s}$, for which τ_C differs by only about $+2.8\%$, while τ_R remains about 45% smaller.

Several limitations help explain these differences. First, oscilloscope loading and the additional input resistance, together with parasitic effects from the switch and wiring, and contact resistance or leakage paths, can add parallel resistive components that are not captured in the simple model. Second, the fit assumes a single exponential, whereas parasitic capacitances in the circuit and oscilloscope could introduce additional time constants. Finally, the $V_R(t)$ data set is incomplete: for unknown reasons the exported trace contains a large gap in the middle of the decay, so the fit to V_R is constrained mainly by the earliest and latest points. This is most likely due to the acquisition or export settings rather than to any physical effect. The missing region, however, reduces our ability to test the model using the resistor voltage and makes the V_R fit less reliable than the V_C fit.

4 Exercise 2

In Exercise 2 I investigated the fast transient response of three different series circuits driven by a square wave from the function generator: an RC circuit, an RL circuit, and an LC circuit. In each case the manual switch and battery of Experiment 1 were replaced by the signal generator, so that each rising or falling edge of the square wave acts like a rapid step in the applied voltage.

For every circuit, the oscilloscope was used to record the voltage across a single component (the resistor in the RC and RL circuits, and either the capacitor or inductor in the LC circuit) during one switching event. The raw traces contained many repeated cycles, but for analysis I deliberately selected a short time window around a single rising or falling edge in which the input can be treated as a single DC step. Representative scope traces are shown in the figures below.

The analysis for this is common to all three circuits. Starting from Kirchhoff's laws and the differential equations in the lab manual, the transient response of the RC and RL circuits is expected to follow a single-exponential decay or rise with a characteristic time constant $\tau = RC$ or $\tau = L/R$, while the LC circuit (with small damping) should show a damped sinusoidal oscillation with an exponential envelope. For each dataset I therefore fit the appropriate theoretical time-domain model (single exponential for RC and RL, damped sinusoid for LC), extract the best-fit time constant τ , and compare it with the value predicted from the measured component values. Chi-squared statistics and residual plots are used throughout to quantify the goodness of fit and to identify any systematic deviations from the simple models.

The uncertainties used in all three analyses are obtained in a consistent way. Voltage uncertainties combine the oscilloscope's vertical gain and offset accuracy with its 8-bit quantization error; time uncertainties come from the time-base accuracy and sampling; and these are propagated into

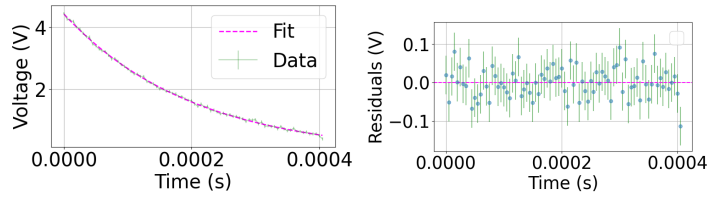


Figure 3: Left) Measured voltage response of the RC circuit in Experiment 2 following a square-wave transition, fitted with the exponential decay model. The best-fit time constant is $\tau = (5.89 \pm 0.03) \times 10^{-4}$ s. The fit yields a chi-squared statistic of $\chi^2 = 60.9$, corresponding to a reduced chi-squared of $\chi^2_v = 0.771$ and a chi-squared probability $p = 0.934$. **Right)** Residuals of the fit with uncertainty bars

the fitted parameters using the standard χ^2 formalism. As in Experiment 1, the oscilloscope input acts as a $1 \text{ M}\Omega \parallel 16 \text{ pF}$ load across whatever component is being measured. This parallel loading slightly reduces the effective resistance seen by the capacitor in the RC circuit and is explicitly included when comparing the fitted RC time constant to theory, while for the RL and LC circuits the series resistances are only of order 10^2 – $10^3 \Omega$, so the additional $1 \text{ M}\Omega$ branch changes the time constants by less than 0.1% and can be neglected in practice.

4.1 RC

We fit $V_R(t)$ with the exponential decay function 18, and compare the fitted time constant τ with the value predicted by the measured R and C .

The best-fit model shown in Fig.3 has a time constant

$$\tau_{RC} = (5.89 \pm 0.03) \times 10^{-4} \text{ s} \equiv (0.589 \pm 0.003) \text{ ms}.$$

The fit yields a chi-squared statistic $\chi^2 = 60.94$ with a reduced chi-squared $\chi^2_v = 0.771$ and a chi-squared probability of $p = 0.934$. A reduced χ^2 slightly below one and a large p -value indicate that the exponential decay model is consistent with the data and that the voltage uncertainties are a bit conservative.

The residuals (lower panel of Fig. 3) are randomly scattered about zero, which supports the adequacy of fitting the single falling edge.

Using the measured component values, the ideal RC time constant would be $\tau_{RC} = RC \approx 6.10 \times 10^{-4}$ s. However, the scope's $1 \text{ M}\Omega$ input is in parallel with R_3 . The effective resistance seen by the capacitor is therefore

$$R_{\parallel} = \left(\frac{1}{R_3} + \frac{1}{1 \text{ M}\Omega} \right)^{-1} \approx 27.3 \text{ k}\Omega,$$

which predicts a loaded time constant $\tau_{RC, \text{loaded}} = 0.594$ ms. The fitted value $\tau_{RC, \text{fit}} = 0.588$ is only about 0.8% smaller than $\tau_{RC, \text{loaded}}$, well within the combined uncertainties from the component tolerances and the fit. Thus, once the $1 \text{ M}\Omega$ oscilloscope load is included, the measured time constant is in very good agreement with the theoretical expectation.

Overall, the RC data show the expected exponential decay behaviour with a single, well-defined time constant. The statistics are very good and residuals support the RC model.

4.2 LR

$V_R(t)$ is fitted with this exponential rise model 16 and we compare the fitted τ with the expected ratio L/R derived from the components.

The best-fit parameters in Fig. 4 give a time constant

$$\tau_{RL, \text{fit}} = (23.6 \pm 0.2) \mu\text{s} \equiv (2.36 \pm 0.02) \times 10^{-5} \text{ s},$$

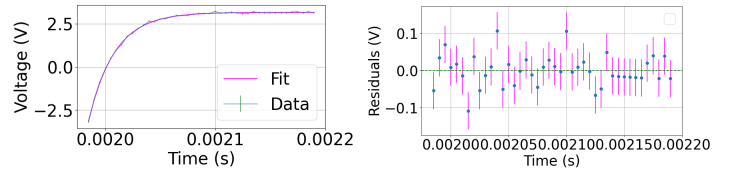


Figure 4: Left) Measured voltage response of the LR circuit following a square-wave transition, fitted with the exponential rise model. The best-fit time constant is $\tau = (28.6 \pm 0.2) \mu\text{s}$. The fit yields a chi-squared statistic of $\chi^2 = 29.1$, corresponding to a reduced chi-squared of $\chi^2_v = 0.77$ and a chi-squared probability $p = 0.849$. **Right)** Residuals of the fit with uncertainty bars.

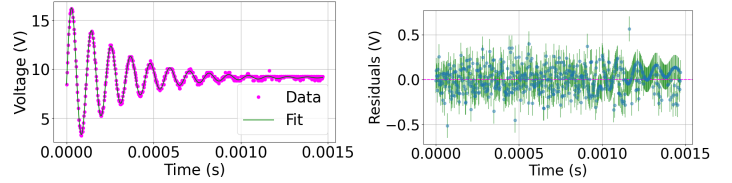


Figure 5: Left) Measured capacitor voltage V_C in the LC circuit following a square-wave transition, fitted with the damped oscillation model. The best-fit decay time constant is $\tau = (7.6 \pm 0.3) \times 10^{-5}$ s. The fit yields a chi-squared statistic of $\chi^2 = 801$, corresponding to a reduced chi-squared of $\chi^2_v = 1.38$ and a chi-squared probability $p = 5.3 \times 10^{-9}$. **Right)** Residuals of the fit with uncertainty bars

with $\chi^2 = 29.1$, $\chi^2_v = 0.77$, and a chi-squared probability $p = 0.849$. The reduced χ^2 closely below one and the large p -value indicate an good statistical agreement with the exponential model, and the residuals (lower panel of Fig. 4) fluctuate symmetrically about zero.

Using the measured component values, the RL time constant would be

$$\tau_{RL} = \frac{L}{R_2} 59.5 \mu\text{s}.$$

This is about two times larger than $\tau = 28.6 \mu\text{s}$ from the fit. Since $\tau = L/R_{\text{tot}}$, the fitted value corresponds to an effective series resistance $R_{\text{tot}} = \frac{L}{\tau_{RL, \text{fit}}} \approx 1.27 \times 10^3 \Omega$. This is significantly larger than the resistor $2 R_2 = 505 \Omega$. The difference can possibly be attributed to the inductor's internal resistance, the output resistance of the function generator, and additional small series resistances in the wiring and contacts, all of which contribute to R_{tot} but not to R_2 .

In this configuration the oscilloscope's $1 \text{ M}\Omega$ input is connected in parallel with R_2 . The combined resistance $R_{2 \parallel \text{scope}} \approx 505 \Omega$ differs from R_2 by less than 0.1%, so the scope loading has a negligible effect on the time constant. The shorter-than-expected τ therefore likely reflects extra resistance in the real RL circuit rather than the oscilloscope input or user error.

Physically, the RL circuit relaxes faster than we would predict from R_2 alone because energy in the inductor is dissipated not just in R_2 , but also in the resistance of the coil, the function generator, and the wiring.

4.3 LC

Capacitor voltage $V_C(t)$ The damped sinusoidal 17 fit to the capacitor data in Fig.6 gives

$$\tau_{LC, C} = (7.6 \pm 0.3) \times 10^{-5} \text{ s} \equiv (76 \pm 3) \mu\text{s},$$

with $\chi^2 = 801$, $\chi^2_v = 1.38$, and a chi-squared probability $p = 5.3 \times 10^{-9}$. Although χ^2_v is close to 1, the tiny p -value indicates that, given the small error bars, the deviations from the simple damped-sinusoid model are statistically significant. This is also visible in the residuals, which show slight oscillatory structure rather than perfectly random scatter.

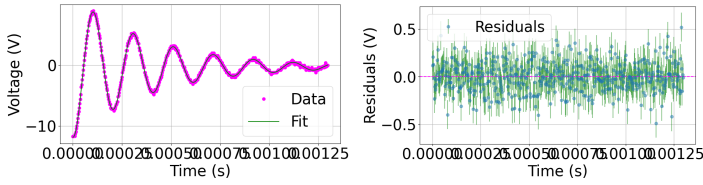


Figure 6: *Left* Measured inductor voltage V_L in an LC circuit following a square-wave transition, fitted with the damped oscillation model. The best-fit decay time constant is $\tau = (4.06 \pm 0.03) \times 10^{-4}$ s. The fit yields a chi-squared statistic of $\chi^2 = 650$, corresponding to a reduced chi-squared of $\chi^2_v = 1.26$ and a chi-squared probability $p = 4.67$. *Right*) Residuals of the fit with uncertainty bars

Inductor voltage $V_L(t)$. For the inductor data in Fig.5 the same functional form yields a best-fit time constant

$$\tau_{LC,L} = (4.06 \pm 0.03) \times 10^{-4} \text{ s} \equiv (406 \pm 3) \mu\text{s},$$

with $\chi^2 = 650.0$, $\chi^2_v = 1.26$, and a small chi-squared probability ($p \ll 1$). Again the reduced χ^2 is close to 1 but the tiny p -value reflects the fact that our uncertainties are small enough that subtle systematic differences between the model and data become significant. The residuals in Fig.5 show subtle patterns sitting on top of otherwise random noise.

In an ideal underdamped series RLC circuit the envelope of all voltages decays with the same time constant $\tau_{RLC} = \frac{2L}{R_{\text{tot}}}$, in principle $\tau_{LC,C}$ and $\tau_{LC,L}$ should be equal.

Instead, I find that the inductor voltage appears to decay over a time-scale about five times longer than that of the capacitor, $\tau_{LC,L} \approx 5\tau_{LC,C}$. Interpreting these as effective damping constants gives $R_{\text{tot},C} \approx 7.9 \times 10^2 \Omega$, $R_{\text{tot},L} \approx 1.5 \times 10^2 \Omega$, which is not consistent with a single series resistance. This discrepancy, together with the structured residuals, shows that our simple single- τ model does not capture all aspects of the real LC transient.

The oscilloscope's 1 M Ω input again provides an additional damping. When measuring V_C , the scope is effectively a 1 M Ω resistor in parallel with the capacitor. This parallel resistor by itself would produce a decay time-scale $R_{\text{scope}}C_{\text{fast}} \approx 0.022$ s, which is more than two hundred times longer than the measured 10^{-4} s envelopes, so it contributes only a very weak extra loss. The differences between $\tau_{LC,C}$ and $\tau_{LC,L}$ arise from other non-idealities: parasitic elements in the circuit, sensitivity of the fit to the precise time window, or even human error.

Even with these issues, the fits clearly show the qualitative behaviour expected for an underdamped RLC circuit: both the capacitor and inductor voltages oscillate with a frequency of a few kilohertz (consistent with the natural resonance frequency $f_0 = (2\pi\sqrt{LC_{\text{fast}}})^{-1} \approx 6$ kHz) while their amplitudes decay exponentially over many cycles. Thus, in all three cases, taking only a short section of the full square wave around one transition was essential: it allowed the data to be compared directly to the transient solutions of the equations.

5 Exercise 3

In Experiment 3 we investigated the frequency-dependent impedance of two simple series circuits, one containing a resistor and capacitor (RC) and the other a resistor and inductor (RL), driven by a sinusoidal wave generator with zero DC offset and 6 V_{pp}. In both cases a known series resistor was used together with a “device of interest” (the capacitor or inductor) so that the current and total impedance could be inferred from oscilloscope measurements of the voltages in the circuit.

For each circuit we collected the driving frequency, the amplitudes of the relevant voltages, and the phase difference between them using the Keysight oscilloscope's Measure functions. In the RC configuration, Resistor 2 3 was in series with

Capacitor 2 3 and we measured the total voltage across the RC combination and the voltage across the resistor, so that the current could be inferred directly from Ohm's law. In the RL configuration the same resistor was placed in series with Inductor 13, but we chose to measure the voltage across the inductor instead of the resistor and inferred the current from the relation $|V_L| = \omega L|I|$.

Our analysis is the same for both circuits. From the measured voltages we first compute the magnitude of the circuit impedance $|Z(\omega)|$ and the phase shift $\phi(\omega)$ between the applied voltage and the current as functions of angular frequency ω . We then fit these data with the theoretical impedance models: for the RC circuit we use the series-RC formulas $|Z_{RC}(\omega)|$ and $\phi_{RC}(\omega)$, and for the RL circuit we use the series-RL formulas $|Z_{RL}(\omega)|$ and $\phi_{RL}(\omega)$. In each case we extract effective component values (a capacitance C_ϕ from the RC phase data, a resistance R_Z from the RC impedance data, and analogous parameters L_ϕ and L_Z, R_Z for the RL circuit), and we use χ^2 statistics and residual plots to quantify how well the simple series models describe the measurements. Finally, we compare the fitted parameters with the independently measured component values and discuss any systematic deviations from the ideal behaviour.

The uncertainties for both the RC and RL analyses are obtained in a consistent way. For each voltage measurement we combine the oscilloscope's vertical gain and offset accuracy with its 8-bit quantization error to obtain a conservative uncertainty on the amplitude; phase are taken from the instrument specifications and combined in quadrature with LSB. When we propagate these uncertainties into $|Z(\omega)|$ and $\phi(\omega)$, they appear as the error bars in all fit and residual plots. We also account qualitatively for the finite input impedance of the oscilloscope, which acts as a 1 M Ω || 16 pF load on whatever node is being measured. This parallel loading is most relevant for the RC circuit at low frequencies, where the series impedance can approach 1 M Ω and thus be slightly reduced by the scope input. In contrast, for the RL circuit the total series impedance remains 1 k Ω over our frequency range, so the oscilloscope's 1 M Ω input changes the effective impedance by at most $\sim 0.1\%$, which is negligible compared to our other uncertainties.

5.1 RC

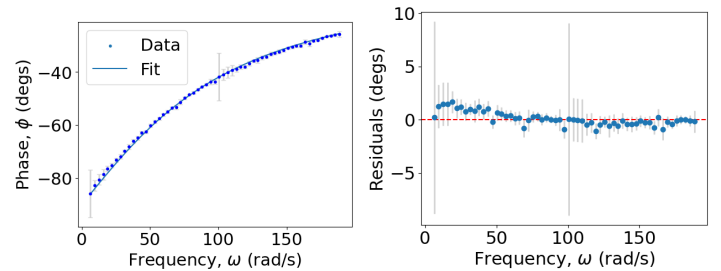


Figure 7: *Left*) Measured phase response of the series RC circuit as a function of driving angular frequency, fitted with the theoretical phase model evaluated in degrees with the resistance fixed at $R = 520 \Omega$. The best-fit capacitance is $C = (21.34 \pm 0.01)$ nF. The fit yields a chi-squared statistic of $\chi^2 = 35.9$, corresponding to a reduced chi-squared of $\chi^2_v = 0.619$ and a chi-squared probability $p = 0.989$. *Right*) Residuals of the fit with uncertainty bars

We expect that the the magnitude of the impedance decreases with increasing frequency, while the phase shifts from -90° at low frequency towards 0° at high frequency. The best-fit capacitance is $C_\phi = (21.34 \pm 0.01)$ nF. The fit yielded a chi-squared statistic $\chi^2 = 35.9$, a reduced chi-squared $\chi^2_v = 0.620$, and a chi-squared probability $p = 0.989$. A reduced χ^2 slightly below 1 and a very high p -value indicate that the phase model is consistent with the data and that the phase uncertainties may be slightly overestimated.

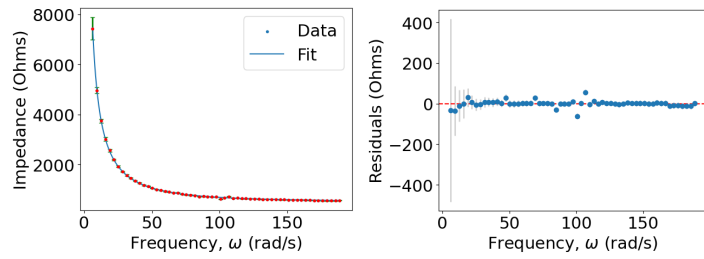


Figure 8: *Left*) Measured magnitude of the impedance $|Z|$ of the series RC circuit as a function of driving angular frequency, fitted with the theoretical RC impedance model. The best-fit resistance is $R = (513 \pm 2) \Omega$. The fit yields a chi-squared statistic of $\chi^2 = 72.6$, corresponding to a reduced chi-squared of $\chi_v^2 = 1.25$ and a chi-squared probability $p = 0.0933$. *Right*) Residuals of the fit.

The residuals in the lower panel of Fig. 7 are scattered symmetrically about zero with no obvious dependence on frequency, which suggests that a simple series RC model with a single capacitance describes the phase response well over the full frequency range. There is no systematic trend that would indicate frequency-dependent capacitance or an incorrect form of the phase function. Comparing the capacitance from the fit with the measured reading $C \approx 21.8$ nF, I find a fractional difference of about 1.9%. Given that C_ϕ incorporates any small non-idealities in the capacitor (e.g., equivalent series resistance and leakage) as well as possible systematic errors in the frequency calibration and phase measurement, this level of agreement is reasonable very good. The close match between the fitted curve and the data across many points supports the interpretation of C_ϕ as a good effective capacitance for the RC impedance model.

The magnitude of the impedance and its fit are shown in Fig. 8. Here I fixed the capacitance in the model to measured value and allowed the resistance to vary. The theoretical curve was fit to the measured $|Z(\omega)|$ values using the theoretical RC impedance model 20. The best-fit resistance is $R_Z = (513 \pm 2) \Omega$. The fit yields $\chi^2 = 72.6$, a reduced chi-squared $\chi_v^2 = 1.25$, and a chi-squared probability $p = 0.0933$. The reduced χ^2 slightly larger than one and the moderate p -value shows that the simple model is mostly adequate but that the deviations are large enough to be noticeable.

The residuals (lower panel of Fig. 8) are mostly clustered near zero but show small systematic deviations at the lowest frequencies, where the measured $|Z|$ is slightly smaller than the model prediction. At low ω the capacitive reactance $1/(\omega C)$ becomes large and the total impedance of the series RC circuit can reach values comparable to the oscilloscope's 1 M Ω input resistance. Since the scope measures the total voltage across the RC circuit, that 1 M Ω input acts as a parallel branch that reduces the effective impedance seen by the source, which naturally leads to measured $|Z|$ values that are a bit lower than the ideal series-RC prediction. This is a plausible explanation for the low-frequency structure in the residuals.

Comparing R_Z with the DC measurement $R_2 = 505 \Omega$, the fitted resistance is about 1.6% larger. This excess can be attributed to additional series resistance from the transformer windings, leads, and the function generator output, which all contribute to the effective series resistance that determines the AC impedance but are not included in the bench reading of R_2 . The fact that R_Z lies between R_2 and the $R_{\text{eff}} = 520 \Omega$ used in the phase fit is consistent with this picture.

Taken together, the phase and magnitude analyses show that the frequency dependence of the series RC circuit is very well described by the manual's impedance formulas when real component values are used. The phase response follows $\phi_{\text{RC}}(\omega) = \arctan[-1/(\omega RC)]$, smoothly evolving from nearly -90° at low frequency (capacitor-dominated behaviour) towards 0° at high frequency (resistor-dominated

behaviour), exactly as in the theoretical RC curve shown in Fig. 4(b)1 of the manual. The magnitude response likewise transitions from $|Z| \approx 1/(\omega C)$ at low frequency to $|Z| \approx R$ at high frequency, matching the qualitative expectations for an RC circuit.

5.2 LR

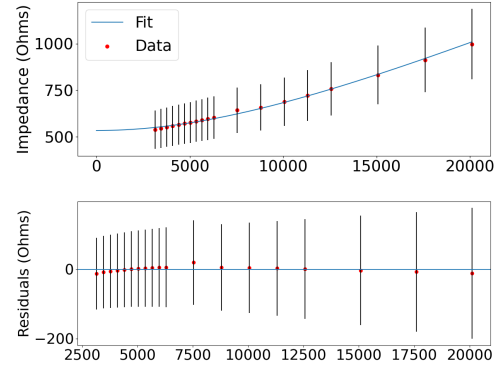


Figure 9: *Left*) Measured magnitude of the ratio of $|Z|$ of the series RL circuit as a function of driving angular frequency, fitted with the theoretical RL impedance model. The best-fit Inductance is $L = (43 \pm 2)$ mH and the best-fit Resistance is $R = (533 \pm 9) \Omega$. The fit yields a reduced chi-squared of $\chi_v^2 = 0.00396$ and a chi-squared probability $p = 1.0$. *Right*) Residuals of the fit with uncertainty bars.

The impedance data and best-fit curve are shown in Fig. 16. Fitting the RL impedance model in Eq. 22 to $|Z(\omega)|$ gives $L_Z = (43 \pm 2)$ mH, $R_Z = (530 \pm 10) \Omega$.

The fit yields a small reduced chi-squared, $\chi_v^2 = 0.004$, with an associated chi-squared probability $p = 1.0$. A reduced χ^2 far below one together with such a high p -value indicates that the scatter of the data is much smaller than implied by the assigned uncertainties; in other words, our error bars are conservatively overestimated. This is consistent with the fact that the oscilloscope specifications used to construct the voltage and phase uncertainties are “worst-case” bounds rather than typical random errors. The residuals in the lower panel of Fig. 16 are scattered about zero with no clear systematic trend, albeit with large uncertainty bars at all frequencies. At low ω the impedance is dominated by the series resistance, while at high ω the inductive reactance ωL becomes large and $|Z|$ increases roughly linearly with frequency. The data follow this expected behaviour closely, supporting the interpretation of L_Z and R_Z as effective parameters describing the series RL circuit.

Comparing R_Z with the DC measurements, the resistor alone has a nominal value $R_2 = 505 \Omega$, and the inductor's DC resistance is $R_L \approx 38.1 \Omega$, giving a total measured series resistance of $R_{\text{tot}} = R_2 + R_L \approx 543 \Omega$. The fitted resistance $R_Z = (533 \pm 9) \Omega$ is therefore about 1.5% smaller than R_{tot} , but this difference lies well within the quoted uncertainty and is consistent with small systematic effects in how we infer the current from the inductor voltage. The fitted inductance $L_Z = (43 \pm 2)$ mH agrees, within uncertainties, with the bench measurement of $L \approx 41$ mH for Inductor 1. Taken together, the impedance fit suggests that the simple series RL model provides an excellent description of the amplitude response of the circuit.

Having determined an effective series resistance R_Z from the impedance data, we used this value as R_{eff} in the phase model of Eq. (23) and fit the measured phase shift $\phi(\omega)$ between the total voltage and the inductor voltage. The phase data and best-fit curve are shown in Fig. 14. The free parameters in this fit were the inductance L and a constant offset b representing any systematic phase error due to cable and instrument delays. The best-fit inductance inferred from the phase data is $L_\phi = (22 \pm 4)$ mH.

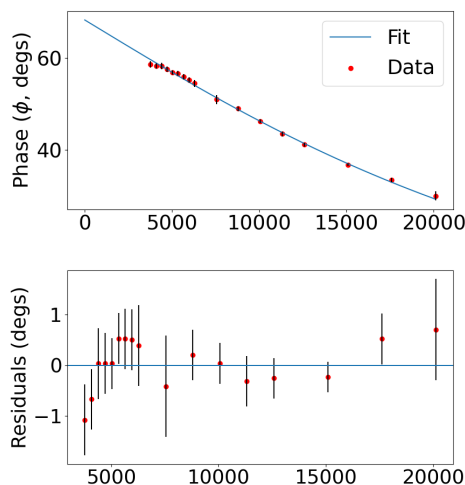


Figure 10: *Left*) Measured phase response of the series RL circuit as a function of driving angular frequency, fitted with the theoretical RL phase model evaluated in degrees with the resistance fixed to the value inferred from the impedance fit. The best-fit inductance from the phase data is $L_\phi = (22 \pm 4)$ mH. The fit yields a reduced chi-squared of $\chi^2_v = 0.652$ and a chi-squared probability $p = 0.833$. *Right*) Residuals of the fit with uncertainty bars.

The fit yields a reduced chi-squared of $\chi^2_v = 0.652$, with a chi-squared probability $p = 0.833$. A reduced χ^2 moderately below one and a large p -value indicate that the RL phase model is statistically consistent with the data and that, as for the impedance analysis, our phase uncertainties are likely somewhat overestimated.

The residuals in the lower panel of Fig. 14 are generally scattered symmetrically about zero with no strong frequency dependence. There is a hint of gentle curvature, but no clear pattern that would definitively signal a failure of the simple RL phase function over the truncated frequency range. At low ω the inductive reactance ωL is small and the measured phases deviate most strongly from the ideal model; these low-frequency points were therefore excluded from the fit in order to avoid biasing the inferred inductance. Over the remaining range the phase decreases smoothly from about 70° towards 30° as the frequency increases, qualitatively consistent with the expected evolution of an RL phase response measured across the inductor. However, the inductance extracted from the phase fit, $L_\phi = (22 \pm 4)$ mH, is significantly smaller than both the impedance-based value $L_Z = (43 \pm 2)$ mH and the nominal inductor value of ≈ 41 mH. This factor of two discrepancy suggests that the phase data is more susceptible to systematic errors than the amplitude data. Measuring the phase difference between V and V_L rather than between V and the resistor voltage means that any non-ideal behaviour of the inductor directly affects the phase relation used in the fit. In addition, small timing offsets of only a few degrees can translate into large fractional changes in the inferred L_ϕ for an RL circuit. We therefore regard L_Z from the impedance analysis as the more reliable estimate of the inductance, while L_ϕ should be interpreted as an effective inductance limited by systematic phase uncertainties.

Overall, the phase and magnitude analyses together show that the frequency dependence of the series RL circuit is largely consistent with the theoretical impedance formulas when realistic component values are used. The magnitude of the impedance increases from a resistance-dominated region at low frequency to an inductive region where $|Z| \approx \omega L$, while the phase shift evolves from nearly 90° (current lagging the applied voltage by almost a quarter cycle) toward 0° as the resistive contribution becomes more important. The good agreement of L_Z and R_Z with the independently measured component values, along with the behaviour of the residuals, indicates that the simple series RL model provides

a reasonable description of the circuit, with the main limitations arising from systematic uncertainties in the phase measurement and possibly human error.

6 Conclusion

Across these three exercises I used time and frequency domain measurements to test the transient and steady-state behaviour of RC, RL and LC circuits against carefully chosen models. In Exercise 1, exponential fits to $V_C(t)$ and $V_R(t)$ during a slow RC discharge showed that, once the oscilloscope's $1\text{ M}\Omega$ loading is included, the capacitor data give a time constant consistent with RC (good χ^2 and residuals), whereas the resistor data yield a much shorter τ and structured residuals, indicating offsets, leakage paths and other non-idealities. In Exercise 2, fitting single exponentials and damped sinusoids to fast RC, RL and LC transients gave an RC time constant in excellent agreement with the loaded prediction, an RL time constant implying a much larger effective series resistance than R_2 alone, and two very different LC decay constants with low p -values, showing that inductor losses, parasitic elements and model simplicity limit the accuracy of our envelopes. In Exercise 3, impedance and phase fits for the RC circuit produced C and R values within a few percent of the independently measured components and largely random residuals, while the RL impedance fit agreed well with the measured L and R but the RL phase fit gave an inductance about a factor of two smaller, revealing that phase measurements across the inductor are possibly more vulnerable to non-ideal inductor behaviour and human error. Overall, the experiments confirm the qualitative predictions of the RC/RL/LC models but highlight the importance of scope loading, parasitic resistances, and careful model choice; future measurements could be improved by measuring extra resistances directly, choosing time windows more carefully, treating the scopes input impedance in our schematic and calculations from the start instead of correcting afterwards, and reduce parasitics by using shorter leads and keeping the circuit compact.

References

- [1] C. Lee, T. Vahabi, E. Horsley, C. Dale, A. Harlick, R. M. Serbanescu, F. Michael Bartram *Currents in LCR*, PHY324 Laboratory Manual, Department of Physics, University of Toronto, current revision 2023.
- [2] Keysight Technologies, *InfiniiVision 1000 X-Series Oscilloscopes – Data Sheet*, 2018–2020.
- [3] Wavetek Meterman, *Model 37XR Autoranging Digital Multimeter – Catalog Data Sheet*, RS Components stock no. 450–5278.

Appendix

Measured Component Values

Capacitor 1	$(1.05 \pm 0.04) \mu F$
Capacitor 2	$(21.8 \pm 0.8) nF$
Inductor 1	$(30 \pm 2) mH$
Inductor 2	$(41 \pm 2) mH$
Resistor 1	$(465 \pm 3) k\Omega$
Resistor 2	$(505 \pm 3) \Omega$
Resistor 3	$(28.0 \pm 0.2) k\Omega$

Table 3: Table of Measured Component Values

Circuit Diagrams

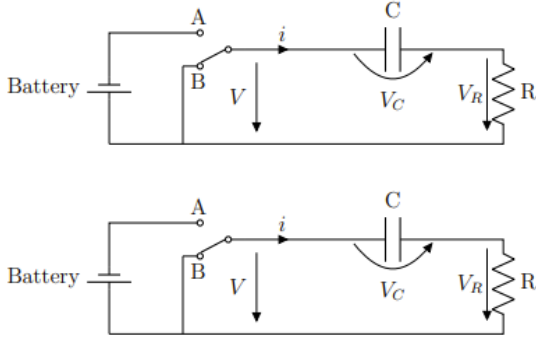


Figure 11: Exercise 1 Setup: Circuit with a battery (far left) supplying a DC current and a switch (top left), Capacitor (C) and resistor (R) in series, and an oscilloscope (channels 1 and 2) to measure both V and V_C respectively.

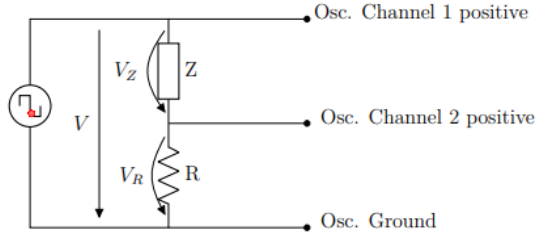


Figure 12: Experimental Setup for Exercise 2. Circuit with a signal generator (far left) supplying a square wave current (simulating a dc switch that switches on and off more quickly), a device (Z) (either capacitor or inductor depending on what is being tested) and resistor (R) in series, and an oscilloscope (channels 1 and 2) to measure both V and V_R respectively.

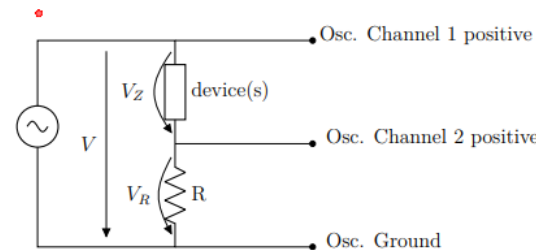


Figure 13: Experimental Setup for Exercise 3. Circuit with a signal generator (far left) supplying a sin wave AC current, a device (Z) (either capacitor or inductor depending on what is being tested) and resistor (R) in series, and an oscilloscope (channels 1 and 2) to measure both V and V_R respectively.

Equations:

$$V_R(t) = A(1 - e^{-t/\tau}) + V_0. \quad (16)$$

$$V(t) = A e^{-t/\tau} \cos(\omega t + \phi) + V_0, \quad (17)$$

$$V_R(t) = A e^{-t/\tau} + V_0 \quad (18)$$

$$V(t) = A e^{-t/\tau} + V_\infty, \quad (19)$$

$$|Z_{RC}(\omega)| = \sqrt{R^2 + \frac{1}{(\omega C_{\text{fast}})^2}} \quad (20)$$

$$\phi_{RC}(\omega) = \arctan\left(-\frac{1}{\omega R_{\text{eff}} C}\right) \quad (21)$$

$$|Z_{RL}(\omega)| = \sqrt{R^2 + \omega L} \quad (22)$$

$$\phi_{RL}(\omega) = b + \arctan(R/\omega L) \quad (23)$$

Plots

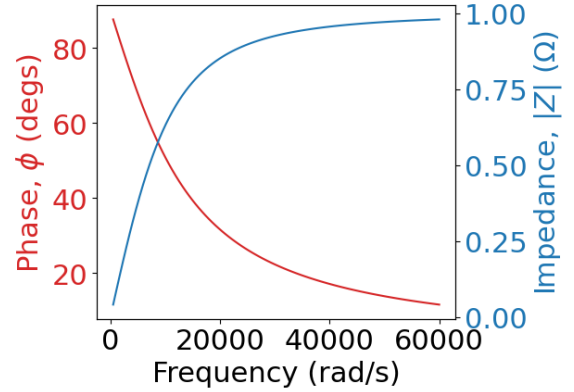


Figure 14: Theoretical plot for the phase and impedance of a series LR circuit measured across Inductor 1

Figures:

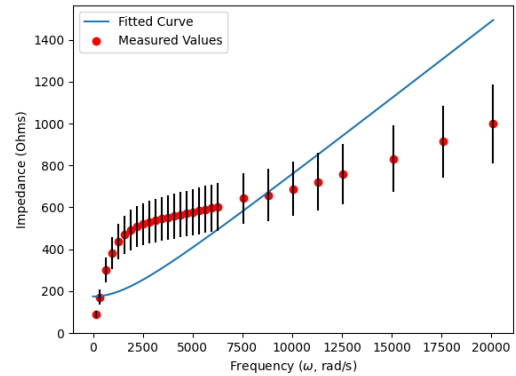


Figure 15: The RL impedance graph from experiment 3 without truncation

Statement of AI Usage only autocomplete from Vscode and the table of equations in the introduction, no other AI usage

Contribution Statement

Mohamed: Intro, Method, References, Ex3 (LR) analysis, Appendix

Julian: Abstract, Title, Analysis, Conclusion

I Meterrman 37XR Multimeter Uncertainty

For the Meterrman 37XR multimeter, uncertainty depends on the measurement mode:

$$\sigma_{\text{reading}} = \begin{cases} 0.001 \times \text{reading} + 5 \times \text{sig} & \text{for voltage (V)} \\ 0.005 \times \text{reading} + 4 \times \text{sig} & \text{for resistance (R)} \\ 0.05 \times \text{reading} + 30 \times \text{sig} & \text{for inductance (I)} \\ 0.03 \times \text{reading} + 5 \times \text{sig} & \text{for capacitance (C)} \end{cases} \quad (24)$$

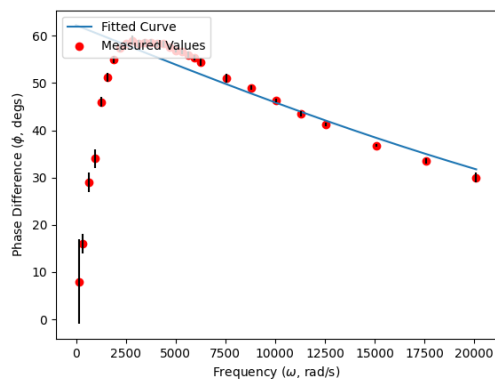


Figure 16: *The RL phase shift graph from experiment 3 without truncation*